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compact1, compact2, compact3

java.time.zone

Class ZoneOffsetTransition

java.lang.Object
    java.time.zone.ZoneOffsetTransition

All Implemented Interfaces:

`Serializable, Comparable<ZoneOffsetTransition>`

```
public final class ZoneOffsetTransition
extends Object
implements Comparable<ZoneOffsetTransition>, Serializable
```

A transition between two offsets caused by a discontinuity in the local time-line.

A transition between two offsets is normally the result of a daylight savings cutover. The discontinuity is normally a gap in spring and an overlap in autumn. `ZoneOffsetTransition` models the transition between the two offsets.

Gaps occur where there are local date-times that simply do not exist. An example would be when the offset changes from `+03:00` to `+04:00`. This might be described as 'the clocks will move forward one hour tonight at 1am'.

Overlaps occur where there are local date-times that exist twice. An example would be when the offset changes from `+04:00` to `+03:00`. This might be described as 'the clocks will move back one hour tonight at 2am'.

Implementation Requirements:

`This class is immutable and thread-safe.`

Since:

`1.8`

See Also:

`Serialized Form`

Method Summary

Concrete Methods

Modifier and Type	Method	Descriptio
int	compareTo(ZoneOffsetTransition transition)	Compares this transition to another based on the transition instant.
boolean	equals(Object other)	Checks if this objec equals another.
LocalDateTime	getDateTimeAfter()	Gets the local transition date-time, as would be expressed with the 'after' offset.
LocalDateTime	getDateTimeBefore()	Gets the local transition date-time, as would be expressed with the 'before' offset.
Duration	getDuration()	Gets the duration of the transition
Instant	getInstant()	

		Gets the transition instant.
ZoneOffset	getOffsetAfter()	Gets the offset after the transition
ZoneOffset	getOffsetBefore()	Gets the offset before the transition
int	hashCode()	Returns a suitable hash code.
boolean	isGap()	Does this transition represent a gap in the local time-line.
boolean	isOverlap()	Does this transition represent an overlap in the local time-line.
boolean	isValidOffset(ZoneOffset offset)	Checks if the specified offset is valid during this transition
static ZoneOffsetTransition	of(LocalDateTime transition, ZoneOffset offsetBefore, ZoneOffset offsetAfter)	Obtains an instance defining a transition between

		two offsets.
long	**toEpochSecond**()	Gets the transition instant as an epoch second.
String	**toString**()	Returns a string describing this object.

Methods inherited from class java.lang.Object

clone, finalize, getClass, notify, notifyAll, wait, wait, wait

Method Detail

of

```
public static ZoneOffsetTransition of(LocalDateTime transition,
                                      ZoneOffset offsetBefore,
                                      ZoneOffset offsetAfter)
```

Obtains an instance defining a transition between two offsets.

Applications should normally obtain an instance from ZoneRules. This factory is only intended for use when creating ZoneRules.

Parameters:

transition - the transition date-time at the transition, which never actually occurs, expressed local to the before offset, not null

offsetBefore - the offset before the transition, not null

offsetAfter - the offset at and after the transition, not null

Returns:

the transition, not null

Throws:

IllegalArgumentException - if offsetBefore and offsetAfter are equal, or transition.getNano() returns non-zero value

getInstant

```
public Instant getInstant()
```

Gets the transition instant.

This is the instant of the discontinuity, which is defined as the first instant that the 'after' offset applies.

The methods `getInstant()`, `getDateTimeBefore()` and `getDateTimeAfter()` all represent the same instant.

Returns:

the transition instant, not null

toEpochSecond

```
public long toEpochSecond()
```

Gets the transition instant as an epoch second.

Returns:

the transition epoch second

getDateTimeBefore

```
public LocalDateTime getDateTimeBefore()
```

Gets the local transition date-time, as would be expressed with the 'before' offset.

This is the date-time where the discontinuity begins expressed with the 'before' offset. At this instant, the 'after' offset is actually used, therefore the combination of this date-time and the 'before' offset will never occur.

The combination of the 'before' date-time and offset represents the same instant as the 'after' date-time and offset.

Returns:

the transition date-time expressed with the before offset, not null

getDateTimeAfter

```
public LocalDateTime getDateTimeAfter()
```

Gets the local transition date-time, as would be expressed with the 'after' offset.

This
is the first date-time after the discontinuity, when the new offset applies.

The combination of the 'before' date-time and offset represents the same instant as the 'after' date-time and offset.

Returns:

`the transition date-time expressed with the after offset, not null`

getOffsetBefore

```
public ZoneOffset getOffsetBefore()
```

Gets the offset before the transition.

This is the offset in use before the instant of the transition.

Returns:

`the offset before the transition, not null`

getOffsetAfter

```
public ZoneOffset getOffsetAfter()
```

Gets the offset after the transition.

This is the offset in use on and after the instant of the transition.

Returns:

`the offset after the transition, not null`

getDuration

```
public Duration getDuration()
```

Gets the duration of the transition.

In most cases, the transition duration is one hour, however this is not always the case. The duration will be positive for a gap and negative for an overlap. Time-zones are second-based, so the nanosecond part of the duration will be zero.

Returns:

`the duration of the transition, positive for gaps, negative for overlaps`

isGap

```
public boolean isGap()
```

Does this transition represent a gap in the local time-line.

Gaps occur where there are local date-times that simply do not exist. An example would be when the offset changes from `+01:00` to `+02:00`. This might be described as 'the clocks will move forward one hour tonight at 1am'.

Returns:

`true if this transition is a gap, false if it is an overlap`

isOverlap

```
public boolean isOverlap()
```

Does this transition represent an overlap in the local time-line.

Overlaps occur where there are local date-times that exist twice. An example would be when the offset changes from `+02:00` to `+01:00`. This might be described as 'the clocks will move back one hour tonight at 2am'.

Returns:

`true if this transition is an overlap, false if it is a gap`

isValidOffset

```
public boolean isValidOffset(ZoneOffset offset)
```

Checks if the specified offset is valid during this transition.

This checks to see if the given offset will be valid at some point in the transition. A gap will always return false. An overlap will return true if the offset is either the before or after offset.

Parameters:

`offset - the offset to check, null returns false`

Returns:

`true if the offset is valid during the transition`

compareTo

`public int compareTo(ZoneOffsetTransition transition)`

Compares this transition to another based on the transition instant.

This compares the instants of each transition. The offsets are ignored, making this order inconsistent with equals.

Specified by:

`compareTo in interface Comparable<ZoneOffsetTransition>`

Parameters:

`transition - the transition to compare to, not null`

Returns:

`the comparator value, negative if less, positive if greater`

equals

`public boolean equals(Object other)`

Checks if this object equals another.

The entire state of the object is compared.

Overrides:

`equals in class Object`

Parameters:

`other - the other object to compare to, null returns false`

Returns:

`true if equal`

See Also:

`Object.hashCode(), HashMap`

hashCode

```
public int hashCode()
```

Returns a suitable hash code.

Overrides:

`hashCode` in class `Object`

Returns:

the hash code

See Also:

`Object.equals(java.lang.Object)`, `System.identityHashCode(java.lang.Object)`

toString

```
public String toString()
```

Returns a string describing this object.

Overrides:

`toString` in class `Object`

Returns:

a string for debugging, not null

Submit a bug or feature
For further API reference and developer documentation, see Java SE Documentation. That documentation contains more detailed, developer-targeted descriptions, with conceptual overviews, definitions of terms, workarounds, and working code examples.
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